

Dual Microphone Preamplifier For ADC, MCU And Arduino

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Most microphones' output signals are weak and cannot be directly connected to some other circuits. Also, some microphones require DC power supply or high impedance inputs. Here is a dual (stereo) preamplifier for microphones that can be used with a wide variety of circuits, including analogue-to-digital converters (ADC), microcontroller (MCU) kits and single-board computers such as Arduino.

Circuit and working

The circuit diagram of a dual microphone preamplifier is shown in

Fig. 1. It is built around two electret microphones (MIC1 and MIC2), two low-voltage MCP602 operational amplifiers (IC1 and IC2), three LEDs (LED1 through LED3), two BC337 transistors (T1 and T2), two 1N4007 diodes (D1 and D2), and a few other components.

The circuit has two independent channels. The first audio channel comprises MIC1, IC1, and T1, and the second channel comprises MIC2, IC2, and T2. CON1 and CON2 are connectors for MIC1 and MIC2, respectively.

Switches S1 and S2 are closed only if microphones need an external

power supply. If small loudspeakers or headphones are used as microphones, these switches should be opened because these devices produce a much lower signal when used as dynamic microphones.

IC1 and IC2 work as non-inverting amplifiers with a gain of 31 each. When switches S3 and S4 are opened, the gain is 31, and when these are closed, the gain is 11. The gain is selected according to the microphones, power supply, and required output levels.

First and second channel outputs are available across connectors CON3

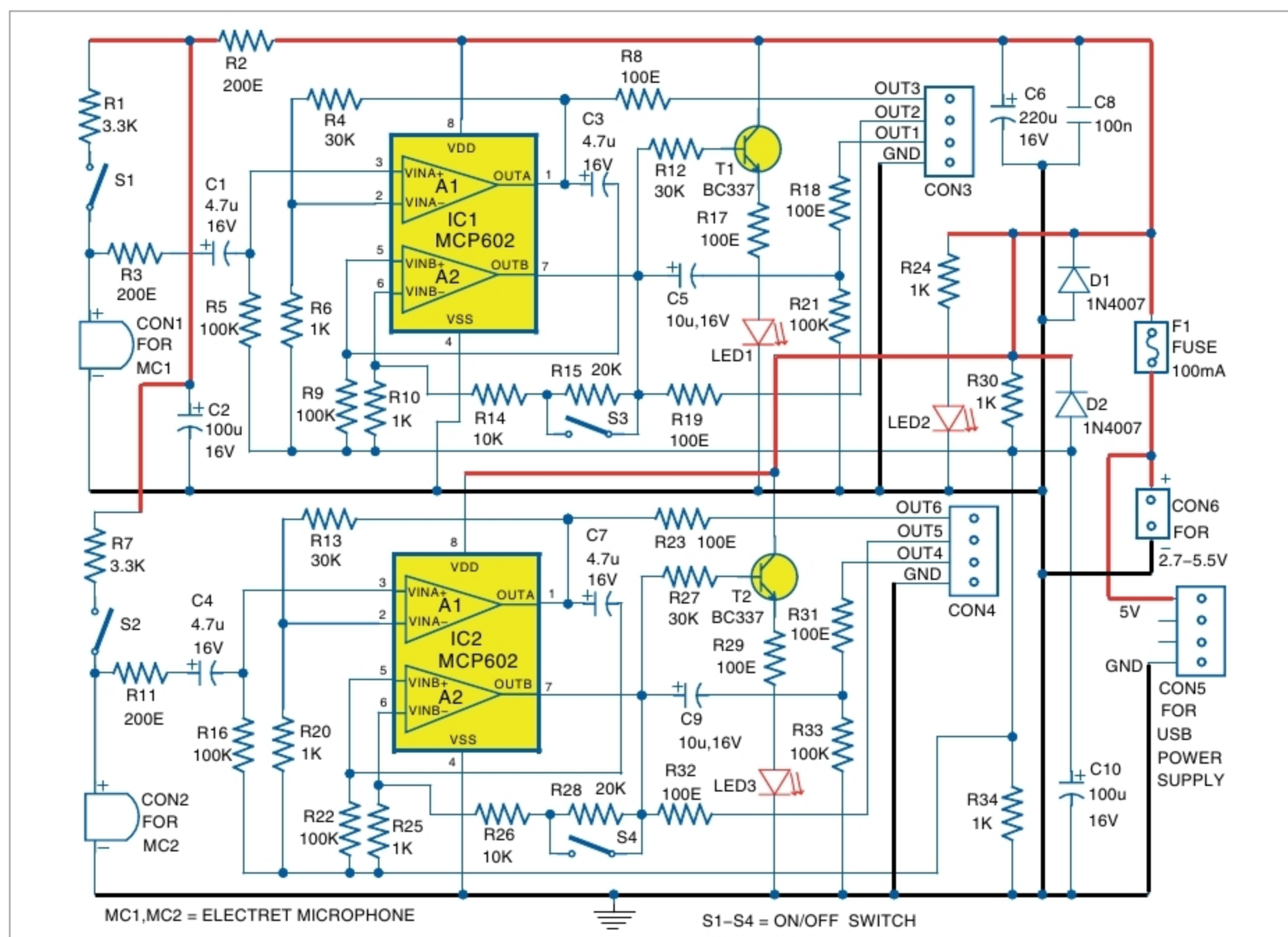


Fig. 1: Circuit diagram of the dual microphone preamplifier